increase risk and aggregate costs, while reducing resilience.[759] Another commenter said that the complexity and costs created by competing consolidators could exceed their benefits to investors.[760] One commenter said that the proposed decentralized consolidation model and the proposed changes to consolidated market data "could introduce significant additional costs, confusion and complexity into an already complex system for equity market data, and raises a number of questions and issues."[761]

The markets currently have a decentralized model of data dissemination with regard to the exchange proprietary data feeds. This decentralized model operates alongside the current centralized consolidation model, and market participants must navigate the different data offerings, connectivity options, and fees. Therefore, the Commission does not believe that a decentralized consolidation model would necessarily increase complexity. Rather, the Commission believes that the decentralized consolidation model may lessen some of the complexities that exist today by eliminating the need to purchase both SIP data and proprietary data for some market participants. Additionally, because the Commission expects that there will be multiple competing consolidators providing consolidated market data products to market participants, rather than one exclusive SIP—the single point of failure that exists in the current model—the Commission believes that the decentralized consolidation model will enhance, rather

---

759 See TechNet Letter II at 1–2. This commenter said that every new competing consolidator brings "exponentially increasing risks" and that the competing consolidator model lacked strong industry support. Id. at 2.

760 See STANY Letter II at 8.

761 FINRA Letter at 3.

than harm, the resiliency of the national market system.[762] If one competing consolidator ceases operations, its impact on the markets should be minimized due to the presence of other competing consolidators that can perform the same functions[763] and the ability of new entrants to serve as competing consolidators.

8. Comments on Surveillance and Regulation in the Decentralized Consolidation Model

Several commenters expressed concern about the proposed decentralized consolidation model's impact on regulation.[764] Some commenters stated that the proposed model would present regulatory risks.[765] Additionally, commenters asked questions related to the proposed model's regulatory impact.[766]

---

762 See infra Section III.C.2. The competing consolidator model is designed to result in multiple viable sources of consolidated market data, not a single source of such data. Therefore, there will not be a single point of failure. However, the second prong of the definition of "critical SCI systems" in Regulation SCI, 17 CFR 242.1000 et seq.—a catch-all for systems that "[p]rovide functionality to the securities markets for which the availability of alternatives is significantly limited or nonexistent and without which there would be a material impact on fair and orderly markets"—would apply in the event that availability of alternatives were significantly limited or nonexistent in the future. See infra Section III.F for a discussion of the application of Regulation SCI to competing consolidators.

763 See infra Section V.C.2(c)(iv) (discussing the benefits of the decentralized consolidation model to market resiliency).

764 See Nasdaq Letter IV at 2, 3, 4, 12–13, 35; TechNet Letter II at 2; Kubitz Letter at 1; Joint CRO Letter at 2, 3, 4; FINRA Letter at 3, 4–5, 6; Citadel Letter at 5; TD Ameritrade Letter at 13.

765 See Nasdaq Letter IV at 35; TechNet Letter II at 2; Kubitz Letter at 1.

766 See Citadel Letter at 5; FINRA Letter at 4, 5, 6; TD Ameritrade Letter at 13.

One commenter stated that the proposal overlooked the effect of the proposed decentralized consolidation model on market surveillance and enforcement.[767] The commenter also said that the proposal would limit exchange market data revenue because revenues would be based upon "some unspecified measure of cost," which would impact exchanges' abilities to perform their self-regulatory functions,[768] while also increasing the cost of regulatory compliance.[769]

Another commenter, representatives from two exchanges, stated that the proposal leaves unclear whether SROs will be required to purchase all of the consolidated market data feeds of competing consolidators and self-aggregators for surveillance purposes, or if SROs can use a limited number of such feeds instead.[770]

The Commission does not believe that the decentralized consolidation model raises unique regulatory risks, undermines effective SRO surveillance, or imposes burdens on broker-dealers. The U.S. equity markets and the regulatory programs that have been developed to oversee their operation already have experience handling multiple sets of data due to the existence of the exclusive SIPs' feeds and proprietary data feeds. Market participants currently can utilize many data options for different purposes, and the SROs are able to develop surveillance programs to oversee their members. Further, in the current model with both SIP data and proprietary data, the SROs develop their surveillance systems based on the data sets

---

767 See Nasdaq Letter IV at 2.

768 See id. at 37; see also NYSE Letter II at 22.

769 See Nasdaq Letter IV at 35; see also NYSE Letter II at 22.

770 See Joint CRO Letter at 3.

they believe best allow them to perform their regulatory obligations.[771] Broker-dealers using different data sets than those used by their SROs already have to respond to SRO surveillance requests based on the different data used by the SROs. This process will not change in the decentralized consolidation model. Surveillance and regulatory programs that utilize SIP data may have to be updated to utilize a new data source, either from a competing consolidator or based on self-aggregation by the SRO.[772] SROs will not be required to purchase every consolidated feed from all competing consolidators to conduct enforcement or surveillance, just as they are not required today to purchase all consolidated (synthetic NBBO) data products provided by each of the different market data vendors aggregating proprietary feeds.

Furthermore, all competing consolidators will register with the Commission and become regulated entities (if not already SROs) subject to Rule 614 of Regulation NMS and Commission oversight. As required by Rule 614, competing consolidators will provide information about

---

[771] Section 6(b)(1) of the Exchange Act provides that an exchange must be so organized and have the capacity to be able to enforce compliance by its members, and persons associated with its members, with the Exchange Act, the rules and regulations thereunder, and the rules of the exchange. See also Section 15A(b)(2) of the Exchange Act. See, e.g., Securities Exchange Act Release Nos. 74690 (Apr. 9, 2015), 80 FR 20282 (Apr. 15, 2015) (proposed rule change from Nasdaq explaining that Nasdaq uses a real-time surveillance system that uses a "mirrored" version of Nasdaq's "NMS feed," which consumes the Nasdaq Protected Quote Service as well as certain proprietary market data feeds and SIP data); 74967 (May 15, 2015), 80 FR 29127 (May 20, 2015) (proposed rule change from Nasdaq PHLX ("Phlx") stating that Phlx's surveillance similarly relies on a mirrored version of Phlx's NMS feed, which consumes the Phlx Protected Quote Service and certain proprietary market data feeds and SIP data). See also Nasdaq Rule 4759(a) and Nasdaq PSX Rule 3304(a) for a list of the proprietary quotation feeds and SIP feeds used by the respective exchanges for the handling, routing, and execution of orders, as well as for regulatory compliance functions related to those functions.

[772] The Commission has modified the definition of self-aggregator so that SROs would be permitted to be self-aggregators.

their operations through a public Form CC, as well as monthly reports on their performance and other metrics relevant to potential subscribers, such as latency, system up-time, and system issues.[773] Like many of the disclosures made by the exclusive SIPs,[774] the information required pursuant to Rule 614(d) will be publicly available.[775] Competitive forces also will incentivize competing consolidators to operate reliably and with low latency, and in conjunction with Commission oversight, the application of Regulation SCI and the required disclosures and transparency provided by Rule 614, should help to ensure high performance and system integrity.

---

773 See Rule 614(d); see also infra Section III.C.8.

774 The information to be published by competing consolidators is based upon information that is currently produced by the CTA/CQ SIP and the Nasdaq UTP SIP, either for public or internal distribution. The exclusive SIPs currently publish to their respective websites monthly processor metrics that provide the following information: system availability, message rate and capacity statistics, and the following latency statistics from the point of receipt by the SIP to dissemination from the SIP: average latency and 10th, 90th and 99th percentile latency. See CTA Metrics, available at https://www.ctaplan.com/metrics (last accessed Nov. 27, 2020); UTP Metrics, available at http://www.utpplan.com/metrics (last accessed Nov. 27, 2020). Additionally, the exclusive SIPs post on their websites any system alerts and the Nasdaq UTP Plan posts vendor alerts as well. See CTA Alerts, available at https://www.ctaplan.com/alerts (last accessed Nov. 27, 2020); UTP-SIP System Alerts, available at http://www.utpplan.com/system_alerts (last accessed Nov. 27, 2020); UTP Vendor Alerts, available at http://www.utpplan.com/vendor_alerts (last accessed Nov. 27, 2020). Further, the exclusive SIPs publish on their websites charts detailing realized latency from the inception of a Participant matching engine event through the point of dissemination from the exclusive SIP. See CTA Latency Charts, available at https://www.ctaplan.com/latency-charts (last accessed Nov. 27, 2020); UTP Realized Latency Charting, available at http://www.utpplan.com/latency_charts (last accessed Nov. 27, 2020).

775 Because this information is useful to current users of the exclusive SIPs and participants of the Equity Data Plans, the Commission believes that it should be made publicly available by competing consolidators.

In response to the comment that stated the proposal would limit exchange market data revenue, hurting exchanges' abilities to perform their self-regulatory functions[776] and increasing the cost of regulatory compliance,[777] the Commission notes that SROs will develop fees for the data content underlying consolidated market data via the effective national market system plan(s), and the SROs will receive their revenue allocation for their data, as they do today.[778] One commenter suggested having an "authority or agency" evaluate whether the proposed decentralized consolidation model could be gamed, arbitraged, or fraudulently used in a way to interfere with retail investors' access to market data and execution of trades.[779] The Commission believes that the adopted rules, as well as the oversight of SROs, competing consolidators, and self-aggregators should help to ensure that retail investors' access to consolidated market data would not be impacted by the decentralized consolidation model. Specifically, Rule 603(b) requires the SROs to provide their information to competing consolidators, which would be responsible for disseminating consolidated market data products to their subscribers, which would likely include broker-dealers that have retail customers. Among other requirements, Rule 603(c) would continue to apply; therefore, broker-dealers and SIPs[780] must continue to provide a

---

776 See Nasdaq Letter IV at 37; see also NYSE Letter II at 22.

777 See Nasdaq Letter IV at 35; see also NYSE Letter II at 22.

778 See infra Section III.E.2. The Commission discusses the potential economic effects of the proposal on exchange proprietary market data revenue in Section V.C.4(a). See infra Section V.C.4(a); see also infra text accompanying notes 2468–2469.

779 Kubitz Letter at 1.

780 Under Rule 600(b)(16) of Regulation NMS, a competing consolidator is defined as a SIP. See infra Section III.C.1(b).

consolidated display of information in the context in which a trading decision can be implemented.

Further, Rule 614(d)(3) requires competing consolidators to make available consolidated market data products to subscribers on terms that are not unreasonably discriminatory.[781] Competing consolidators will also be subject to the requirements of Rule 614(d), which, among other things, mandate the filing of a public Form CC to provide operational transparency, as well as the public monthly disclosure of metrics and other information concerning performance and operations. These requirements should allow subscribers of a competing consolidator to verify that consolidated market data products are provided in a prompt, accurate, and reliable manner and thus motivate competing consolidators to continue to provide consolidated market data products accordingly.

Commenters also raised questions related to the proposed decentralized consolidation model's regulatory impact.[782] One commenter asked whether there would be any regulatory immunity differences between competing consolidator offerings from SROs and non-SROs.[783] SRO immunity considerations would depend on the particular facts and circumstances.[784]

---

781 See infra Section III.C.8.

782 See Citadel Letter at 5; FINRA Letter at 4–5, 6.

783 See Citadel Letter at 5.

784 See also infra Section III.C.7(a)(iv); Brief of the Securities and Exchange Commission, Amicus Curiae, No. 15-3057, City of Providence v. Bats Global Markets, Inc. (2d Cir.), at 21, 22. Courts have found that SROs are entitled to absolute immunity from private claims under certain circumstances. In particular, "when acting in its capacity as a SRO, [the SRO] is entitled to immunity from suit when it engages in conduct consistent with the quasi-governmental powers delegated to it pursuant to the Exchange Act and the regulations and rules promulgated thereunder." See DL Capital Group, LLC v. NASDAQ Stock Market, Inc., 409 F. 3d 93, 97 (2d Cir. 2005) (quoting D'Alessio v. New York Stock Exchange, Inc., 258 F. 3d 93, 106 (2d Cir. 2001)). If an SRO fails to comply with the provisions of the Exchange Act, the rules or regulations thereunder, or its own rules, the Commission is authorized to take action. See 15 U.S.C. 78s(g).

9. Access to Data: Rule 603(b)

(a) Proposal

Rule 603(b) of Regulation NMS currently requires a centralized consolidation model. The Commission proposed to amend Rule 603(b) to require each SRO to provide its NMS information, including all data necessary to generate consolidated market data, to all competing consolidators and self-aggregators in the same manner and using the same methods, including all methods of access and data formats, as such SRO makes available any information to any other person. Under the proposed approach, competing consolidators and self-aggregators would collect each SRO's market data that is necessary to generate consolidated market data.[785] As proposed, the same access options available to proprietary feeds, including, but not limited to, transmission medium (i.e., fiber optics or wireless), multicast communication, co-location options, physical port, logical port, bandwidth, and FPGA services, would be required to be made available for proposed consolidated market data feeds. Further, any enhancements to proprietary feed methods of access would similarly be made to consolidated market feeds.[786]

In the Proposing Release, the Commission stated that the exchanges could utilize their existing proprietary data product offerings that contain consolidated market data elements or the exchanges could develop new consolidated market data offerings for purposes of making information available under Rule 603(b). Competing consolidators and self-aggregators could choose to purchase products that include only the proposed consolidated market data elements or

---

785 See Proposing Release, 85 FR at 16769.

786 See id.

products that contain elements of both proposed consolidated market data and other proprietary data.[787]

The Commission also proposed to remove the requirement in Rule 603(b) that "all consolidated information for an individual NMS stock [be disseminated] through a single plan processor"[788] because it would be inconsistent with the proposed decentralized consolidation model. In a decentralized consolidation model, multiple competing consolidators would disseminate consolidated market data in individual NMS stocks rather than single plan processors.[789]

In the proposal, the Commission stated that the proposed decentralized consolidation model and the proposed consolidated market data definition would not preclude the exchanges from continuing to sell proprietary data and that the fees for proprietary data, which are outside of the proposed definition of consolidated market data, would be subject to the rule filing process pursuant to Section 19(b) of the Exchange Act and Rule 19b-4.[790] The Commission stated that if an exchange provided its proprietary data products to a competing consolidator or self-aggregator and a competing consolidator or self-aggregator developed a product, or otherwise used data, that exceeded the scope of proposed consolidated market data (e.g., full depth of book data), the competing consolidator or self-aggregator would be charged separately for the

---

787 See id.

788 17 CFR 242.603(b).

789 Proposing Release, 85 FR at 16771.

790 15 U.S.C. 78s(b) and 17 CFR 240.19b-4. See id. at 16769.

proprietary data use pursuant to the individual exchange fee schedules.[791] Self-aggregators and competing consolidators that limited their use of exchange data to proposed consolidated market data elements would be charged only for proposed consolidated market data pursuant to the effective national market system plan(s) fee schedules, in accordance with Rule 608 of Regulation NMS.[792]

(b) Final Rule and Response to Comments

The Commission is adopting the amendments to Rule 603(b) as proposed. The Commission believes that these changes to Rule 603(b) are appropriate to establish the decentralized consolidation model. Under Rule 603(b), the SROs are required to make available all quotation and transaction information that is necessary to generate consolidated market data in the same manner and using the same methods, including all methods of access and the same format, as such SRO makes available any information with respect to quotations for and transactions in NMS stocks to any person. Competing consolidators and self-aggregators will be able to collect from the SROs that NMS information that is necessary to generate consolidated market data as defined in Rule 600(b)(19),[793] which includes core data,[794] regulatory data, administrative data, and self-regulatory organization-specific program data.

Under Rule 603(b), the SROs are allowed to provide their core data to competing consolidators and self-aggregators via the existing proprietary data feeds, a combination of

---

791 See id.

792 17 CFR 242.608.

793 See also infra Section III.C.8(a).

794 For example, competing consolidators and self-aggregators will collect quotation information that is necessary to calculate the NBBO as described in Rule 600(b)(50).